PRACTICE: Slope-Intercept Form

y = mx + b

Name: ________________________________ Date: _______________ Period: _______

SLOPE: m = (y2 - y1) / (x2 - x1) = rise / run
SLOPE-INT: y = mx + b (m = slope, b = y-intercept)

Horizontal --> m = 0
Vertical --> m = undefined

① Calculate Slope from Two Points

1. Find the slope: (2, 3) and (5, 9) m = __________

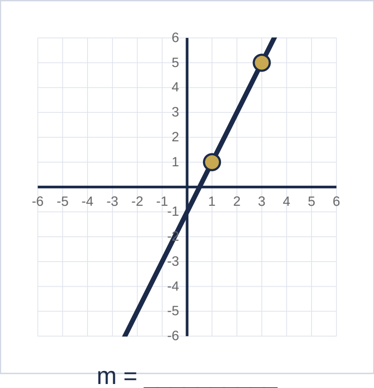
2. Find the slope: (-1, 4) and (3, -2) Write as a simplified fraction! m = __________

3. Find the slope: (4, 1) and (4, 7) m = __________

② Calculate Slope from a Graph

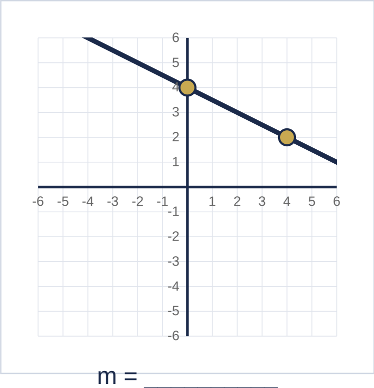
Identify two points on each line. Calculate rise/run.

4.



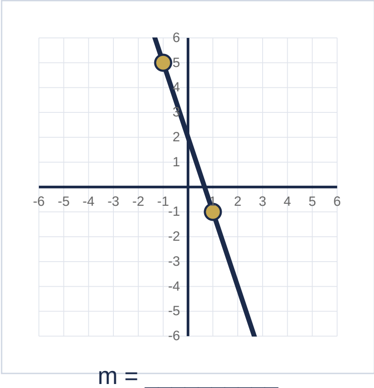
m = __________

5.



m = __________

6.



m = __________

③ Identify Slope & Y-Intercept

7. y = 3x + 5 m = _________ b = _________

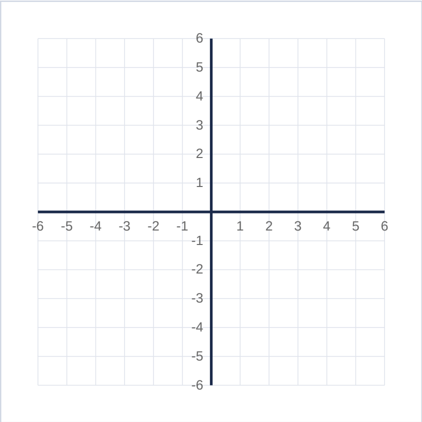
8. y = -1/2 x + 2 m = _________ b = _________

9. y = -4x m = _________ b = _________

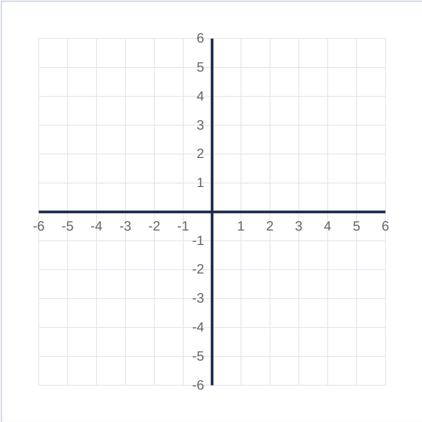
④ Graph Equations in Slope-Intercept Form

Plot the y-intercept. Use slope (rise/run) to find a second point. Draw the line through both.

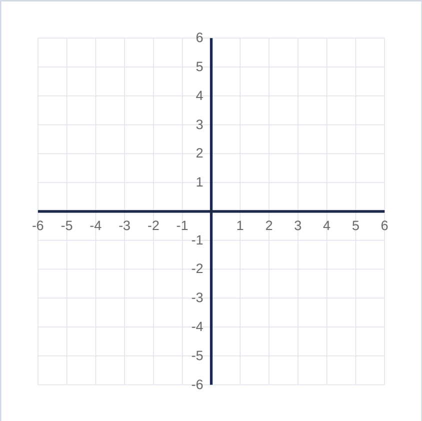
10. $y = 2x - 1$



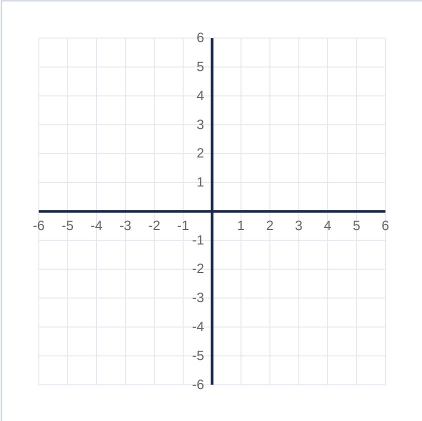
11. $y = -1/2\,x + 3$



12. $y = -x + 4$



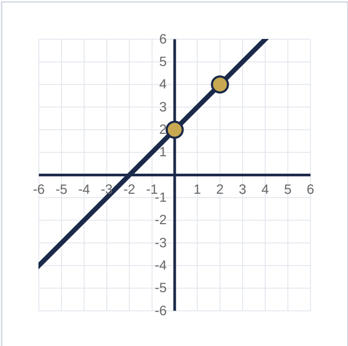
13. $y = 1/3\,x + 1$



⑤ Write Equations from a Graph

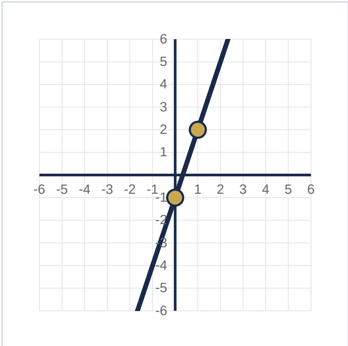
Find the y-intercept and slope. Write the equation in $y = mx + b$ form.

14.



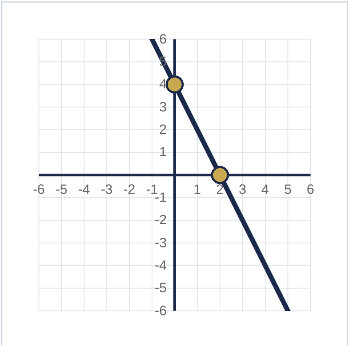
y = ______________________

15.



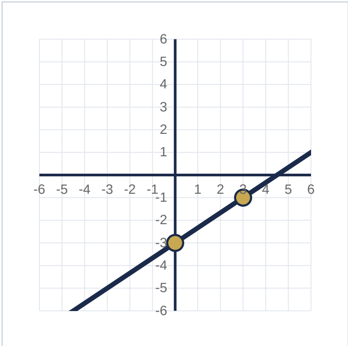
y = ______________________

16.



y = ______________________

17.



y = ______________________